

Abha Singh Sardar

M.Tech Smart Manufacturing

CGPA: 8.40

GitHub: github.com/abhaanisha



ACADEMIC PROFILE

M.Tech. Smart Manufacturing	8.40	Indian Institute of Science, Bangalore	2027
M.Tech. Computer Science and Engineering	8.15	Maulana Azad National Institute of Technology, Bhopal	2017
B.Tech. Computer Science and Engineering	7.91	Kalinga Institute of Industrial Technology	2014
Class XII (ICSE)	75.67%	Church School Beldih	2010
Class X (ICSE)	80.57%	Church School Beldih	2008

INTERNSHIPS

Dept. of Cyber-Physical Systems (RBCCPS), IISc	Research Intern	May'26 - Jul'26
- Preprocessed a multi-subject, student-collected IMU dataset for Human Activity Recognition into clean, windowed training data. - Developed a Human Activity Recognition model on the cleaned data as the full-precision baseline for model compression. - Performed extensive pruning and quantization of the HART transformer model, comparing accuracy, model size and latency for deployment on low-power edge devices.		
Mohanpur Organics Pvt. Ltd.	Analytics & Outbound Automation Intern	May'26 - Jun'26
- Built a Retrieval-Augmented Generation (RAG) chatbot for medical assistance that answers from curated healthcare documents. - Deployed a live demo, CareCompass, on Streamlit (Live Demo GitHub) using hybrid BM25 + MiniLM search over 12 curated WHO, NHS and UNICEF health documents. - Added a rule-based safety check (19 red-flag rules in 3 languages, 5 urgency levels), reaching 100% emergency recall and 3.8% under-triage on a 52-case test set.		

PROJECTS

Head Gesture Recognition System	GitHub	Mar'26 - May'26
- Built a wearable device that turns head gestures into messages for people with limited hand mobility, using ~140K IMU samples (7 classes, 50 Hz) from two Arduino Nicla Vision boards. - Created 148 features (statistical, frequency and left-right symmetry) to separate similar gestures such as Tilt Left and Tilt Right; trained Decision Tree, Random Forest and MLP models. - Deployed an INT8-quantized model (~33.6 KB) on the device with 100% test accuracy and 15-40 ms latency, with live alerts on an OLED screen, a buzzer and caregiver phone notifications.		
FabricaTech - Manufacturing Execution System Interface	GitHub	Jan'26 - May'26
- Designed and built a shop-floor interface in React + TypeScript (Vite, Tailwind CSS) with 7 workflows: login, work orders, quality audit, machine lockout and shift reports. - Built a real-time production dashboard showing OOE, availability, performance and quality yield, with machine data, failure alerts and separate logins for 4 plant roles.		
IMU-Based Human Activity Recognition on Edge	GitHub	Jan'26 - Feb'26
- Built an end-to-end TinyML pipeline on Arduino Nicla Vision to classify 12 activities (e.g., eating, pick-and-place, stair climbing) from 120K+ IMU samples recorded at 50 Hz. - Extracted 72 features (mean, RMS, energy) over 20-sample sliding windows and ran a Decision Tree on the device, reaching 99.43% accuracy on fast activities such as jogging and cycling.		
E-Commerce Recommendation Engine		Jan'26
- Built a collaborative-filtering recommender using Truncated SVD on a sparse 50,000-interaction user-product matrix from a 2,845-product beauty catalogue. - Recommended similar products from latent-factor correlations, returning items with Pearson $r > 0.90$.		
Hybrid CNN-ViT Model for Plant Disease Classification	Research Scholar Springer	May'23 - Aug'23
- Designed a model combining CNN local features with Vision Transformer attention, reaching 98.23% test accuracy on 21,000 tomato-leaf images; published by Springer (CVIP 2023). - Lowered test loss vs. baseline CNNs (0.3182 $\rightarrow$ 0.0956 and 0.2108 $\rightarrow$ 0.0343) and raised F1 scores (0.8842 $\rightarrow$ 0.9819 and 0.9608 $\rightarrow$ 0.9947) on two plant-disease datasets.		
Hybrid Encrypted Reversible Data Hiding	Master's Thesis GitHub	Aug'16 - Jul'17
Developed a reversible data-hiding method combining wavelet transform, SVD and encryption with exact image recovery, achieving >50 dB PSNR and 72% better visual quality than existing methods (MATLAB).		

CERTIFICATIONS		
Google Data Analytics Professional Certificate	Coursera	2024
Python for Data Science, AI & Development	Coursera	2023
DeepLearning.AI TensorFlow Developer	Coursera	2022
Machine Learning	Coursera	2022

SKILLS	
• Programming Languages: Python, MATLAB, TypeScript, MicroPython • AI / ML: Machine Learning, Deep Learning, CNNs, Vision Transformers, NLP, RAG & LLMs, Recommender Systems • Edge AI & TinyML: Pruning, INT8 Quantization, TensorFlow Lite, On-device Inference, IMU Sensing, Arduino Nicla Vision • Libraries & Frameworks: TensorFlow, Keras, scikit-learn, NumPy, Pandas, Sentence-Transformers, React, Tailwind CSS • Tools & Deployment: Git/GitHub, Jupyter, Streamlit, Gradio, Docker, LLM APIs (Groq, Gemini, OpenAI) • Domain: Smart Manufacturing, Manufacturing Execution Systems, Signal Processing, Data Analytics, Scientific Writing	

COURSES	
Basics of Computing, AI and Data Science for Design and Manufacturing, Edge AI, Introduction to Natural Language Processing, Data Mining, Intelligent Mobile Robotics, Sensors and Transducers (PG)	

PUBLICATIONS	
• Ranjan, A., Sardar, A. S., Ranjan, V., & Sharma, N. Improving phishing detection using attention-augmented CNN-GRU-RNN networks. 5th International Conference on Communication, Computing and Electronics Systems (ICCCES 2026). <i>Accepted</i>. • Sardar, A. S., & Ranjan, V. (2023). Enhancing computer vision performance: A hybrid deep learning approach with CNNs and vision transformers. CVIP 2023, Springer, pp. 591–602. DOI: 10.1007/978-3-031-58174-8_49 • Tiwari, N., & Sardar, A. S. (2017). Advancement in reversible data hiding techniques. <i>International Journal of Computer Science and Network Security</i>, 17(3).	

POSITION OF RESPONSIBILITY	
• Placement Representative, M.Tech Smart Manufacturing (2025–27), Dept. of Design & Manufacturing, IISc Jul’26 – Present • Meta Reviewer, IEEE International Students’ Conference SCEECS’25 (Jan 2025); recognised for excellent review work.	

EXTRA-CURRICULAR ACTIVITIES	
• Moderated the Ripples’26 Panel Discussion at IISc Open Day on “<i>Beyond Automation: Artificial Intelligence in Engineering Design and Production</i>”, with speakers from academia and industry. • Hosted the Tech Japan event (India–Nagoya Career Connect, with the City of Nagoya and OCCaP, IISc), connecting IISc students with Japanese industry delegates. • Represented the Department of Design and Manufacturing at HR Connect 2026, interacting with industry HR leaders.	

LANGUAGES	English, Hindi
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